DSA0405

Fundamentals of Data Science

Unit – 3

Case Study Problems

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<u>Q1.</u> Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Case Study 1 – Employee Salaries

Given Data:

4, 5, 5, 6, 7, 8, 8, 9, 30 (₹ lakhs)

Number of observations,

$$n = 9$$

1. Find the Mean, Median and Mode

Mean

Formula:

$$\text{Mean} = \frac{\sum x}{n}$$

Sum of all salaries

$$4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30 = 82$$

Therefore,

$$\text{Mean} = \frac{82}{9} = 9.11$$

Mean = 9.11 lakhs

Median

Arrange the data (already arranged):

4, 5, 5, 6, **7**, 8, 8, 9, 30

Since

$$n = 9$$

Median position

$$= \frac{n + 1}{2} = \frac{9 + 1}{2} = 5$$

5th value = **7**

Median = 7 lakhs

Mode

Number Frequency

4	1
5	2
6	1
7	1
8	2
9	1
30	1

The highest frequency is **2**.

Therefore,

Mode = 5 lakhs and 8 lakhs

(Bimodal data)

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2. Find the Range

Formula

$$\text{Range} = Maximum - Minimum$$

Maximum = 30

Minimum = 4

$$Range = 30 - 4 = 26$$

Range = 26 lakhs

3. Calculate Variance and Standard Deviation

Step 1: Mean

$$\mu = 9.11$$

Step 2: Find $(x - \mu)^2$

x	$x - \mu$	$(x - \mu)^2$
4	-5.11	26.11
5	-4.11	16.89
5	-4.11	16.89
6	-3.11	9.67
7	-2.11	4.45
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.39

$$\sum (x - \mu)^2 = 512.88$$

Variance

Formula

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

$$= \frac{512.88}{9} = 56.99$$

Variance = 56.99

Standard Deviation

Formula

$$\sigma = \sqrt{56.99}$$
$$= 7.55$$

Standard Deviation = 7.55

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4. Identify the Outlier

Most salaries lie between

₹4 lakhs and ₹9 lakhs.

Only **₹30 lakhs** is extremely higher than all others.

Hence,

Outlier = ₹30 lakhs

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5. Which Measure Best Represents the Typical Salary? Explain

Answer: Median

Explanation

- Mean = 9.11 lakhs
- Median = 7 lakhs

The salary ₹30 lakhs is an outlier.

Because of this extreme value,
the mean increases from around ₹7 to ₹9.11 lakhs.

The median remains unaffected.

Hence, the median better represents the typical employee salary.

Q2. Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be right-skewed or left-skewed.
5. Explain why the median may be more useful than the mean.

Case Study 2 – Website Response Time

Given Data

120, 125, 128, 130, 132, 135, 140, 145, 150, 300 (ms)

Number of observations

$$n = 10$$

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1. Calculate Mean and Median

Mean

Formula

$$\text{Mean} = \frac{\sum x}{n}$$

Sum:

$$120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 = 1505$$

Therefore,

$$\text{Mean} = \frac{1505}{10} = 150.5$$

Mean = 150.5 ms

Median

Since there are 10 observations,

Median

$$= \frac{5th + 6th}{2}$$

$$= \frac{132 + 135}{2} = \frac{267}{2} = 133.5$$

Median = 133.5 ms

2. Find the Range

Maximum = 300

Minimum =120

$$Range = 300 - 120 = 180$$

Range =180 ms

3. Calculate Variance and Standard Deviation

Mean

$$\mu = 150.5$$

Table

x	x−150.5	(x−150.5)²
120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25
300	149.5	22350.25

$$\sum (x-\mu)^2 = 25580.5$$

Variance

$$= \frac{25580.5}{10} = 2558.05$$

Variance =2558.05

Standard Deviation

$$= \sqrt{2558.05} = 50.58$$

Standard Deviation =50.58 ms

4. Determine the Skewness

Since one value (300 ms) is much larger than the remaining values,

The data is

Positively Skewed (Right Skewed).

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5. Why is the Median More Useful?

The response time **300 ms** is an outlier.

It increases the mean considerably.

The median is not affected by this value and therefore gives a better estimate of the normal website response time.

Hence,

Median is more useful than the mean.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the **IQR**.
3. Use the **1.5 × IQR rule** to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean.

Q3. Case Study 3 – Student Performance

Given Data:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Number of observations,

$$n = 12$$

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1. Find Q1, Q2 and Q3

Step 1: Arrange the data

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Q2 (Median)

Since $n = 12$(even),

Median

$$= \frac{6^{th} + 7^{th}}{2}$$
$$= \frac{52 + 55}{2} = \frac{107}{2} = 53.5$$

Q2 = 53.5

Q1

Lower half:

42, 45, 48, 50, 52, 52

Q1

$$= \frac{3^{rd} + 4^{th}}{2}$$
$$= \frac{48 + 50}{2} = 49$$

Q1 = 49

Q3

Upper half:

55, 58, 60, 62, 65, 95

Q3

$$= \frac{3^{rd} + 4^{th}}{2}$$
$$= \frac{60 + 62}{2} = 61$$

Q3 = 61

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2. Calculate IQR

Formula

$$IQR = Q_3 - Q_1$$
$$= 61 - 49 = 12$$

IQR = 12

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3. Identify Outliers using 1.5 × IQR Rule

Lower Limit

$$\begin{aligned}&Q_1 - 1.5(IQR)\\&= 49 - 1.5(12)\\&= 49 - 18 = 31\end{aligned}$$

Upper Limit

$$\begin{aligned}&Q_3 + 1.5(IQR)\\&= 61 + 18 = 79\end{aligned}$$

Any value

- Less than 31
- Greater than 79

is an outlier.

Only **95** is greater than 79.

Outlier = 95

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4. Find Mean and Median

Mean

Formula

$$Mean = \frac{\sum x}{n}$$

Sum

$$42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95 = 684$$

$$Mean = \frac{684}{12} = 57$$

Mean = 57

Median

Already calculated

$$Median = 53.5$$

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5. Explain How the Outlier Affects the Mean

The score **95** is much higher than the other marks.

It increases the average marks from the value that most students scored.

- Mean = 57
- Median = 53.5

The mean is affected by the outlier, whereas the median remains almost unchanged.

Therefore, the **median gives a better representation of the typical student performance.**

Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Q4. Case Study 4 – E-Commerce Orders

Given Data:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Number of observations

$$n = 10$$

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1. Calculate Mean, Median and Mode

Mean

Formula

$$Mean = \frac{\sum x}{n}$$

Sum

$$18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35 = 250$$

$$Mean = \frac{250}{10} = 25$$

Mean = 25

Median

Since

$$n = 10$$

Median

$$= \frac{5^{th} + 6^{th}}{2}$$
$$= \frac{24 + 25}{2} = 24.5$$

Median = 24.5

Mode

22 occurs twice.

All other values occur once.

Mode = 22

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2. Calculate Variance and Standard Deviation

Mean

$$\mu = 25$$

x	**x−25**	**(x−25)²**
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0

x	x−25	(x−25)²
26	1	1
28	3	9
30	5	25
35	10	100

$$\sum (x - \mu)^2 = 228$$

Variance

$$\sigma^2 = \frac{228}{10} = 22.8$$

Variance = 22.8

Standard Deviation

$$\sigma = \sqrt{22.8} = 4.77$$

Standard Deviation = 4.77

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3. Calculate Coefficient of Variation

Formula

$$CV = \frac{\sigma}{\mu} \times 100$$
$$= \frac{4.77}{25} \times 100$$
$$= 19.08\%$$

Coefficient of Variation = 19.08%

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4. What Does the Standard Deviation Tell the Company?

The standard deviation of **4.77 orders** means that the daily number of orders usually differs from the average by about **5 orders**.

This helps the company:

- Estimate fluctuations in demand.
- Plan inventory.
- Schedule employees efficiently.

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5. Would You Describe the Order Data as Highly or Moderately Variable?

Since the coefficient of variation is only **19.08%**, the variation is relatively low.

Hence, the data is **moderately variable**, indicating that daily orders are fairly consistent.

Q5. **Case Study 5 — Comparing Two Data Science Models**

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6
Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.
2. Calculate the variance for both models.
3. Calculate the standard deviation for both.
4. Which model has more consistent predictions?
5. Can the model with the lower mean error necessarily be considered better? Explain.

Case Study 5 – Comparing Two Data Science Models

Model A

2, 3, 3, 4, 4, 5, 5, 6

Model B

1, 1, 2, 4, 5, 7, 8, 9

Number of observations

$$n = 8$$

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1. Calculate the Mean Error

Model A

Sum

$$2 + 3 + 3 + 4 + 4 + 5 + 5 + 6 = 32$$

$$Mean = \frac{32}{8} = 4$$

Mean (Model A) = 4

Model B

Sum

$$1 + 1 + 2 + 4 + 5 + 7 + 8 + 9 = 37$$
$$Mean = \frac{37}{8} = 4.625$$

Mean (Model B) = 4.625

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2. Calculate the Variance

Model A

Mean = 4

x	**x−4**	**(x−4)²**
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1

x	x−4	(x−4)²
6	2	4

$$\sum (x-\mu)^2 = 12$$

Variance

$$= \frac{12}{8} = 1.5$$

Variance (Model A) = 1.5

Model B

Mean = 4.625

x	x−4.625	(x−4.625)²
1	-3.625	13.141
1	-3.625	13.141
2	-2.625	6.891
4	-0.625	0.391
5	0.375	0.141
7	2.375	5.641
8	3.375	11.391
9	4.375	19.141

$$\sum (x-\mu)^2 = 69.875$$

Variance

$$= \frac{69.875}{8} = 8.734$$

Variance (Model B) = 8.734

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3. Calculate the Standard Deviation

Model A

$$SD = \sqrt{1.5} = 1.22$$

Standard Deviation = 1.22

Model B

$$SD = \sqrt{8.734} = 2.96$$

Standard Deviation = 2.96

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4. Which Model Has More Consistent Predictions?

Model A has:

- Lower variance (1.5)
- Lower standard deviation (1.22)

Model B has:

- Higher variance (8.734)
- Higher standard deviation (2.96)

Therefore, Model A has more consistent predictions because its prediction errors are closer to the mean.

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5. Can the Model with the Lower Mean Error Necessarily Be Considered Better? Explain

No.

A model with a lower mean error is **not always better** because:

- Mean error measures only the average prediction error.
- Variance and standard deviation measure how consistent the predictions are.
- A model with low mean error but high variability may produce unreliable predictions.

Conclusion: A good model should have both **low mean error** and **low standard deviation** for accurate and consistent performance.